

Machine Learning–Based 100-Year Flood Flow Prediction Model Using Basin Characteristics and Meteorological Data in the Northeast United States

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ABSTRACT: Estimating the value of extreme hydrological events, such as the 100-yr flood at ungauged locations, remains a significant hydrological challenge. Conventional flood flow estimation techniques have significant shortcomings, including random and systematic errors that limit their effectiveness in supporting decision-making in water resource planning. This paper presents the use of random forests (RFs), a machine learning (ML) technique that integrates multiple dynamic and static datasets to estimate the 100-yr flood flow and quantify error in the model predictions. Inputs for the proposed model include precipitation, temperature, slope, watershed area, land cover, and elevation datasets. The 98 gauge locations over the Northeast United States, all with a minimum of 40 years of historic streamflow, are selected to evaluate the ML-based approach that calculates the 100-yr peak flow estimates and compares them to estimates made by the U.S. Geological Survey's streamflow statistics (StreamStats) program. A k -fold cross-validation technique is used to test the flood flow prediction model. The ML technique provides improvements in 100-yr flood estimates, demonstrating similar mean absolute relative error (MARE) to StreamStats but significantly improved normalized centered root-mean-square error (NCRMSE) and Kling–Gupta efficiency (KGE). This estimation approach can support the accurate characterization of error in predicting 100-yr flood, which is essential in the development of flood flow prediction algorithms.

SIGNIFICANCE STATEMENT: The research presents a machine learning–based framework over the Northeast United States and compares its performance to the U.S. Geological Survey (USGS) streamflow statistics (StreamStats)-derived parametric method. The k -fold cross-validation experiment was carried out using ground-reference 100-yr flood flows derived from a network of gauges (98 gauge locations) with an availability of a minimum of 40 years of historic daily streamflow. The cross-validation experiment captured the transferability of the machine learning (ML) technique. We conclude that the proposed flood prediction methodology is robust and accurate given its promising performance in a large study area with heterogeneity in terrain morphology and extreme precipitation. Thus, the machine learning framework can be easily applied to other complex terrain regions, where high-resolution and accurate climate data record is available.

KEYWORDS: Error analysis; Land surface model; Model errors; Model evaluation/performance; Nonlinear models; Regional models

1. Introduction

Floods are the most frequent and devastating natural disaster, causing severe economic damages, increasing human suffering, and destroying infrastructure across the globe (Balica et al. 2013). Extreme flood events have far-reaching and long-lasting impacts on the Northeast United States impacting agriculture, creating property damage, destroying basic infrastructure, negatively impacting fish and wildlife, and decreasing water resources and energy security (Tabari 2020; Wing et al. 2022; Ming et al. 2020; Jessup and Colucci 2012). Climate change is projected to increase extreme precipitation and floods over large parts of the globe (Tabari 2020; Cayan et al. 1999; Karan et al. 2014). These severe events have become more frequent and devastating to economies and livelihoods (Ming et al. 2020). Therefore, accurately estimating the frequency and magnitude of future flood flows is imperative for assessing vulnerability

and flood risk (Haddad and Rahman 2011; Carsell et al. 2004) and developing flood control and mitigation strategies (e.g., Khosravi et al. 2021).

Climate factors (e.g., extreme precipitation events and the timing of snowmelt due to changes in temperatures) contribute to flood vulnerability over the globe (Ahmad et al. 2019; Devi et al. 2019; Saber et al. 2020). Many physical factors, relatively independent of climate change (e.g., soil type, land cover, and other drainage basin characteristics such as slope, elevation, and drainage area), can also increase the severity of floods (Doocy et al. 2013; Hunt 2005; Chakraborty et al. 2005). These factors combine to provide a unique climate–land-use–hydrologic signature, while interactions between the factors can amplify the impacts of hydroclimatic extremes (Pfahl et al. 2017; Tabari 2020; among others). However, it is generally difficult to predict the magnitude and return periods of floods with great accuracy due to the lack of reliable climate data and physical conditions, such as topography, watercourses, and limitations of proper flood prediction methods (Devi et al. 2019; Saber et al. 2020; Pfahl et al. 2017; Tabari 2020).

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Numerous studies have been performed that attempt to estimate extreme events like the 100-yr flood because of the vital role they play in flood risk management (Saleh et al. 2017; Hapuarachchi et al. 2011; Blöschl et al. 2008; Ming et al. 2020). Numerous process-based hydrology models have been applied to estimate extreme flows induced by extreme precipitation (Chiang et al. 2007; Nayak et al. 2005). At a catchment scale, hydrological models are commonly used to estimate extreme events (Blöschl et al. 2008; Li et al. 2017). However, developing, calibrating, and verifying a hydrology model can require significant time and economic and computer resources (Kratzert et al. 2019a; Nearing et al. 2021). The multitude of parameters used in process-based models requires time-consuming calibration to increase the accuracy of the model estimates (Nearing et al. 2021). Also, in an operational flood forecasting system, direct prediction of flood hydrodynamics in real time is still technically complicated (Nearing et al. 2021).

The accelerating development of modern machine learning (ML)-based models has attracted substantial attention as an alternative approach to process-based modeling for water research applications (Kratzert et al. 2019a). Recent studies have shown that ML-based models, calibrated on data from gauged basins, produced more accurate streamflow predictions in ungauged basins than traditional process-based models (Nearing et al. 2021; Kratzert et al. 2019a). ML-based models are efficient in integrating information from diverse datasets under a variety of hydrologic conditions for ungauged basins (Kratzert et al. 2019a). In addition, Kratzert et al. (2019a) showed that regionally trained ML models outperformed several traditional, physically based hydrological models by extracting information from hydrologically diverse basins.

Various recent studies have applied ML methods, such as artificial neural networks (ANNs), support vector machines (SVMs), and random forests (RFs), to predict streamflow events with infrequent return periods (e.g., the 100-yr flood), due to their superior modeling efficiency (Soleymani et al. 2016; Mosavi et al. 2018; Zhang et al. 2016; Kratzert et al. 2019b; Tongal and Booij 2018; DeSanto et al. 2023). Extensive research has been conducted based on the tree-based, nonparametric RF techniques in recent years, and they have been shown to be effective in hydrological classification problems using high-dimensional data (Abraham et al. 2019; Pham et al. 2021; Chen et al. 2020; Tongal and Booij 2018). Moreover, RF techniques have received increasing attention in extreme hydrological event predictions over the rainfall-dominated basins (Tongal and Booij 2018; Mosavi et al. 2018; Papacharalampous and Tyrallis 2018). Specifically, RF identifies the nonlinear relationship between the inputs and outputs and can rapidly adapt to new information (Breiman 2001; Meinshausen and Ridgeway 2006), making them well suited for real-time streamflow prediction (Tongal and Booij 2018) and a good candidate for this study. RF models offer clear interpretations of physical processes through partial dependency plots (PDPs) (Friedman 2001; Crawford et al. 2019; Zhao and Hastie 2021), aligning with human understanding of natural phenomena. RF algorithms utilize an ensemble of several tree predictors such that each tree relies on a random and independent selection of features to create robust predictions (Breiman 2001; Meinshausen and

Ridgeway 2006). Therefore, we will employ one of the most robust approaches, a tree-based nonparametric RF algorithm to estimate the 100-yr flood flow using basin characteristics and multiple meteorological variables in the Northeast United States.

This study evaluates the application of an RF algorithm over the Northeast United States and compares its performance to the U.S. Geological Survey's (USGS) streamflow statistics (StreamStats)-derived parametric method (least squares regression in log space) (Ries et al. 2008). In the Northeast United States, StreamStats is widely applied as a spatial analytical tool to estimate flow statistics for water resource planning and management (Ries et al. 2008). Moreover, StreamStats provides detailed drainage basin characteristic information at any location in both gauged and ungauged basins (Guthrie et al. 2009; Ries et al. 2008). In this study, ground-reference 100-yr flood flows derived from a network of gauges (98 gauge locations) with an availability of a minimum of 40 years of historic daily streamflow are utilized. Estimation of the 100-yr flood flow is used due to its importance in both the United States and globally as a standard for managing flood hazards (<https://www.usgs.gov/special-topics/water-science-school/science/100-year-flood>). The reference value for the 100-yr flood flow data is determined using log-Pearson type III (LPT III) distribution method (Bobee 1975). Several studies utilize LPT III for hydrologic parameter estimation, such as streamflow (Burkey 2022; Hu et al. 2020). This study examines the advantages of the proposed ML technique over process-based approaches and its ability to characterize the estimation uncertainty. A cross-validation experiment that captures the transferability of the RF technique is presented. Particularly, a k -fold validation experiment is performed to assess the transferability of the RF technique.

The specific objectives of this study include the following:

- 1) developing a data blending ML-based technique using 100-yr flood flow estimates over the Northeast United States,
- 2) assessing the ML model using cross-validation experiment,
- 3) evaluating the ML-based and StreamStats-derived regression models by comparing their predicted error characteristics in terms of the error metrics, and
- 4) analyzing the sensitivity of the ML algorithm for the different input variables to present the physical interpretation of the ensemble-based ML model.

2. Data and study area

The study area is the northeastern United States (Maine, New Hampshire, Vermont, Massachusetts, Rhode Island, Connecticut, New York, New Jersey, and Pennsylvania) plus Maryland, Delaware, Virginia, and West Virginia. This region has a wide range of soil types, climate conditions, and elevations. To quantify extreme hydrologic events in this region, the USGS' Hydro-Climatic Data Network (HCDN-2009) (Lins 2012) was used as a source of data to represent the watersheds, varying in size and physical attributes. The HCDN is a subset of USGS streamgages that are essentially unimpaired, suggesting that their flows reflect these prevailing meteorological conditions (Lins 2012).

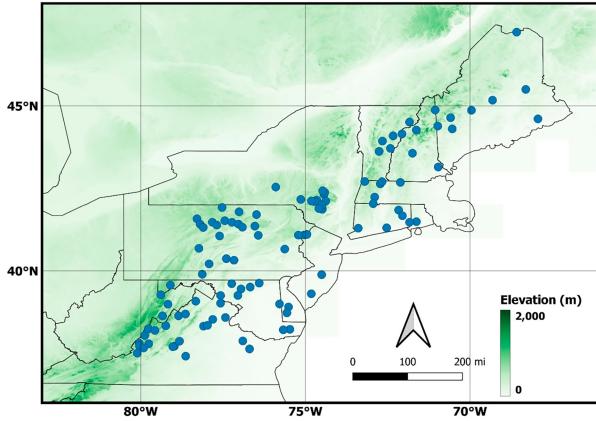


FIG. 1. Geographic locations and terrain elevation map of the study areas. The blue dots represent the streamgauges used.

The United States is classified into smaller hydrologic units that are categorized into four levels: regions, subregions, accounting units, and cataloging units (<https://water.usgs.gov/GIS/huc.html>). Specifically, watersheds are defined by the USGS using a nationwide system based on surface hydrologic features. The USGS defines the hydrologic area with a unique hydrologic unit code (HUC) (<https://nas.er.usgs.gov/hucs.aspx>). HUC 1 and HUC 2 watersheds (Seaber et al. 1987) were selected from the HCDN-2009 set, and that list was further screened to remove gauges with inadequate historic streamflow records. A 40-yr minimum streamflow record was required for consideration. Peak flow varied from $1.73 \text{ m}^3 \text{ s}^{-1}$ ($\sim 61 \text{ ft}^3 \text{ s}^{-1}$) to $2548.51 \text{ m}^3 \text{ s}^{-1}$ ($\sim 90\,000 \text{ ft}^3 \text{ s}^{-1}$), and the drainage area varied between 5.44 km^2 ($\sim 2.1 \text{ mi}^2$) to 3675 km^2 ($\sim 1419 \text{ mi}^2$) for the 98 watersheds (Fig. 1).

Daily maximum precipitation and daily maximum temperature values were extracted for each watershed from the Livneh et al. (2015) hydrometeorological dataset (Livneh and NCAR Staff 2019) during the corresponding streamflow record for each streamgauge. This dataset contains the temperature and precipitation data from approximately 20 000 weather stations in Global Historical Climatology Network (GHCN)-daily (United States), Environment Canada, and Servicio Meteorológico Nacional (Mexico) (Livneh and NCAR Staff 2019). These data are available online, with gridded climate data at $1/16^\circ$ horizontal resolution ($\sim 6 \text{ km}$) from 1950 to 2013 (Livneh and NCAR Staff 2019). If a streamgauge's streamflow record surpassed the year 1950, the full Livneh record (1950–2013) was used to extract the daily maximum precipitation and temperature. Auxiliary variable data such as elevation, slope, land cover, and watershed area were collected from the USGS StreamStats Data-Collection Station report (<https://streamstats.usgs.gov/ss/Views/gagepage.html>).

In summary, the input variables used in the model are 1) watershed area, 2) average elevation of the watershed, 3) average slope of the watershed, 4) percentage of the watershed determined to be forest, 5) percentage of the watershed determined to be wetland, 6) the maximum daily precipitation

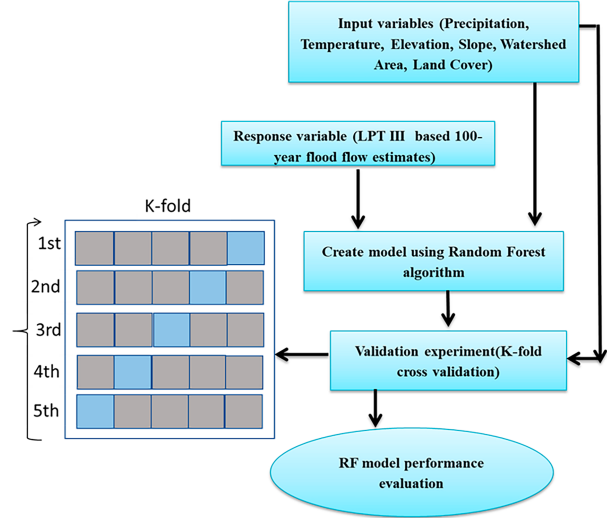


FIG. 2. Schematic representation of the 100-yr flood flow prediction process.

averaged over the watershed, and 7) the maximum daily temperature averaged over the watershed.

3. Methodology

The 100-yr flood flow prediction framework applied here has five primary steps (Fig. 2). Initially, a machine learning-based 100-yr flood flow prediction model was constructed using meteorological and physical variables including precipitation, temperature, slope, watershed area, land cover, and terrain elevation. The reference 100-yr flood flow data were calculated using the LPT III distribution method (Bobee 1975) which is widely applied for such estimates (Hu et al. 2020; Griffis and Stedinger 2007; Burkey 2022). The LPT III distribution was recommended by U.S. federal agencies as a base method for flood frequency analysis (Phine and Hira 1983; Griffis and Stedinger 2007). Specifically, LPT III is a member of the Pearson type III family of distributions, that identifies ranges for the log space skew and the log space standard deviation using the statistical properties of the distribution (Griffis and Stedinger 2007). Moreover, recent studies found that LPT III is more stable than the other flood estimation methods [e.g., generalized extreme value (GEV)] (Hu et al. 2020). It is noted that LPT-III-based flood estimates were determined from available annual maximum flow data in the study area. For the assessment of the ML-based flood flow prediction model, all 98 gauge stations with their corresponding variables were considered. Only those stations with at least 40 years of observations were used to determine the LPT-III-based 100-yr flood estimates. In the ML modeling process, LPT-III-based 100-yr flood estimates were considered the response variable. Inputs to the ML model included two meteorological variables (precipitation and temperature), along with multiple auxiliary variables (slope, watershed area, land cover, and terrain elevation). The ML technique was evaluated using a k -fold validation experiment, and

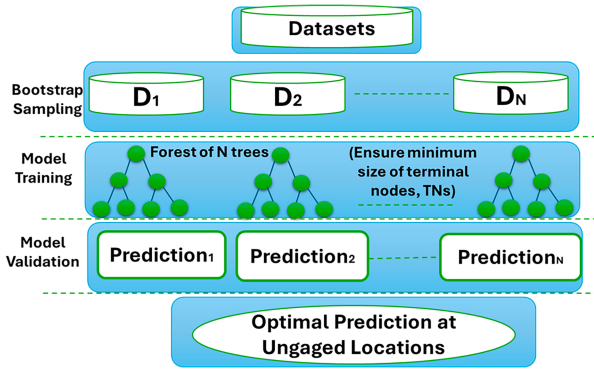


FIG. 3. Schematic representation of the RF-based 100-yr flood flow prediction process for this study.

StreamStats-derived least squares regression was also considered in the study.

a. A random forest data blending algorithm

The ensemble approach of the RF algorithm (Breiman 2001) is relatively robust to outliers (Meinshausen and Ridgeway 2006); hence, its application is suitable in climate and hydrometeorological studies (Pang et al. 2017; Bhuiyan et al. 2019; Crawford et al. 2019). A schematic representation of the random forest framework for this study is presented in Fig. 3. RFs use bootstrap aggregation (or bagging), where bootstrap samples are randomly chosen from a dataset with substitution and a decision tree model is constructed using various subsets of the dataset. This method is known as “out of bag estimation.” This approach avoids “overfitting” by improving the stability and accuracy of the algorithm (Breiman 2001; Meinshausen and Ridgeway 2006). To enhance the model’s efficiency, the individual decision trees are grown deep where each leaf node of the tree has a few training samples. Specifically, RF introduces randomness in the ensemble to reduce the correlation between trees, thus helping to avoid overfitting. Finally, the model estimate is derived by averaging all the regression trees.

The nonparametric, tree-based regression RF model is optimized by tuning its multiple hyperparameters. Hyperparameters in machine learning algorithms are parameters whose values are fixed before the learning process starts. The performance of the RF algorithm depends on three essential hyperparameters: 1) the number of input variables randomly sampled as candidates at each split (“ M_i ”); 2) the number of trees that include the forest (“ N_i ”); and 3) the tree depth, which can be defined by the minimum size of terminal nodes (“ TN_s ”) (Scornet 2017; Coleman et al. 2021).

b. Tuning the RF model algorithm

An RF algorithm is optimized by fitting the training data with different number of trees, of N_i , to minimize and stabilize the out-of-bag error versus the number of trees. The out-of-bag error, commonly used in machine learning models utilizing trees (random forests, boosted decision trees, etc.), is an estimate of the prediction performance by evaluating predictions on the observations that were not used in the building of the next base

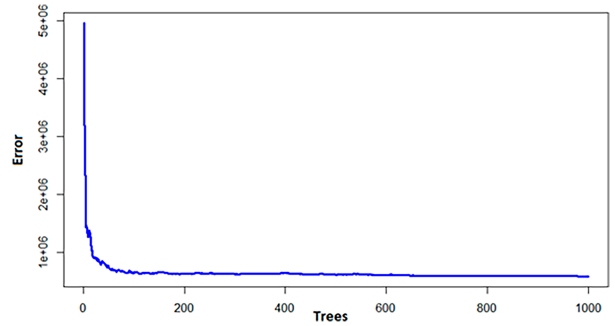


FIG. 4. Out-of-bag error against the number of trees in the RF for the study area.

learner. Additionally, the model’s accuracy greatly depends on the size of the forest. For example, if the number of trees is too small, then the out-of-bag estimation may suffer from high variance. For the RF optimization, we employed a randomized search cross-validation (RandomizedSearchCV) process to obtain optimal hyperparameters and avoid overfitting. In our study, to generalize the RF, the hyperparameters were adjusted for this algorithm combining all the datasets. The major hyperparameters, such as the “ M_i ,” “ N_i ,” and “ TN_s ,” were determined for the RF algorithm through one of the most popular approaches to tune RF hyperparameters called randomized search cross validation (RandomizedSearchCV) using the entire dataset (Bergstra and Bengio 2012). RandomizedSearchCV is a procedure in which random combinations of hyperparameters are chosen to find the best solution to build a machine learning model (Bergstra and Bengio 2012). RandomizedSearchCV is very useful when many hyperparameters are used in the model predictions. Typically, hyperparameters in the RF are fixed before the learning process begins (Meinshausen and Ridgeway 2006; Breiman 2001). Specifically, RandomizedSearchCV uses K -fold cross validation where all the datasets were divided into k numbers of folds (subsamples). The K -fold utilizes entire data for training/validation and also for determining the model prediction error. Overall, k -fold-based RandomizedSearchCV procedure helps to mitigate overfitting and estimates the optimal hyperparameters of the model.

Therefore, N_i is determined for the RF algorithm through a trial-and-error procedure (Scornet 2017; Coleman et al. 2021). The convergence of the error rate, which occurs when the out-of-bag error no longer sharply declines with the addition of more trees (around 100 trees in Fig. 4), heavily depends on the appropriate selection of N_i . A small number of trees will most likely lead to slow convergence of the algorithm, while large values of N_i can lead to convergence failure. During this convergence stage, choosing the other hyperparameters (TN_s and M_i) is frequently performed by trial and error (Scornet 2017; Coleman et al. 2021). Specifically, this process is repeated until the trees reach a specified depth or terminal node size while the curve converges. In this flood flow prediction model, N_i and TN_s are set to 1000 and 5, respectively. Model error converged before the number of trees reached 300 (Fig. 4), where RF utilized the optimal number $TN_s = 5$ for split point selection at each node. Overall, the RF flood flow prediction model

utilized the optimal number N_t , TN_s , and M_t , which controlled the split-variable randomization feature of the algorithm. Since all hyperparameters were finalized in the modeling system, splitting features are consistent for the optimized algorithm.

c. Log-Pearson type III algorithm

The log-Pearson type III distribution has been widely applied due to its accurate estimation of flood frequency analysis (Griffis and Stedinger 2007; Hu et al. 2020). Hu et al. (2020) demonstrated that the LPT III distribution has a low uncertainty in the upper quantiles. The skewness is very sensitive to sample size for LPT III distribution. Therefore, numerous years of streamflow data (a minimum of 30–35 years of historic streamflow) are necessary to obtain accurately the skewness (Hu et al. 2020).

The LPT III distribution is determined using the following equations:

$$\log x = \overline{\log x} + K\sigma_{\log x}, \quad (1)$$

$$C_s = \frac{n \sum (\log x - \overline{\log x})^3}{(n-1)(n-2)(\sigma_{\log x})^3}, \quad (2)$$

where x is the flood flow, $\overline{\log x}$ is the average of the $\log x$ flow values, $\sigma_{\log x}$ is the standard deviation of the $\log x$ values, and K is a frequency factor, which is a function of the return period and skewness coefficient C_s . The n is the record length in years.

d. Cross-validation experiments

A cross-validation experiment was employed to quantitatively assess the performance of the flood flow prediction model. In this analysis, initially, LPT-III-based 100-yr flood flow was evaluated using a k -fold (fivefold) cross-validation experiment in which the dataset was randomly split into five partitions. The model is then trained on the training set, and each holdout sample is treated as a validation dataset. Specifically, all gauge measurements (for the 98 gauge stations) were used to develop the model in the machine learning application. The 5-time random holdout was applied using 80% of the data as training and 20% of the data as validation.

e. Performance evaluation error metrics

Various error metrics were employed to assess the RF-based 100-yr flood flow prediction model, including normalized centered root-mean-square error (NCRMSE), mean absolute relative error (MARE), and the Kling–Gupta efficiency (KGE). NCRMSE was defined as

$$\text{NCRMSE} = \sqrt{\frac{\frac{1}{n} \sum_{i=1}^n \left[\hat{y}_i - y_i - \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) \right]^2}{\frac{1}{n} \sum_{i=1}^n y_i^2}}. \quad (3)$$

Here, y_i is the LPT-III-based 100-yr flood flow (observation), $\hat{y}_i$ is the model-predicted 100-yr flood flow, and n is the

quantity of samples used in the calculation. NCRMSE ranges from 0 (an optimal value) to positive infinity. An NCRMSE value of 0 indicates no random error, while 1 indicates that the random error is equal to 100% of the mean reference streamflow.

MARE was used to measure the systematic error and is defined as the mean of the absolute relative percentage error, calculated by the normalized average:

$$\text{MARE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i} \right|. \quad (4)$$

For an unbiased model, the MARE would be 0.

KGE (Gupta et al. 2009) was used to show the overall flood model performance, which is widely adopted for global hydrological applications (Formetta et al. 2011; Beck et al. 2016). KGE represents the difference between unity and the Euclidian distance (ED) from the ideal point in the three-dimensional criteria space:

$$\text{KGE} = 1 - \text{ED}, \quad (5)$$

$$\text{ED} = \sqrt{(r-1)^2 + (\alpha-1)^2 + (\beta-1)^2}. \quad (6)$$

KGE values range from negative infinity to 1. KGE signifies the agreement between the model predictions and the validation gauge observations, whose best value is 1, to assess the fit of two data pairs. Here, r represents Pearson's correlation coefficient. Moreover, α and β indicate the variability error and the bias error, respectively. The defined parameters are expressed as follows:

$$\alpha = \frac{\sigma_m}{\sigma_o}, \quad (7)$$

$$\beta = \frac{\mu_m}{\mu_o}, \quad (8)$$

where σ_m is the standard deviation of model; σ_o is the standard deviation of reference; μ_m is the mean of the model; and μ_o is the mean of reference.

4. Results and discussion

a. Flood frequency analysis

Flood frequency analyses were used to determine the 100-yr flood flow for the study area. Particularly, the LPT III distribution (Bobee 1975) was applied using annual peak flow observations to determine the 100-yr flood flow. An example of flood frequency estimates (USGS site number: 01181000) based on the LPT III distribution method is shown in Fig. 5, where the resulting curve (Bulletin 17B) was reported together with confidence intervals. The confidence intervals were determined from bootstrapping experiments. The original streamflow record (86 years) for this gauge site was randomly sampled to the derived flood frequency estimates. Similarly, a series of flood frequency analyses were performed for all the gauge stations in our selected hydrological regions.

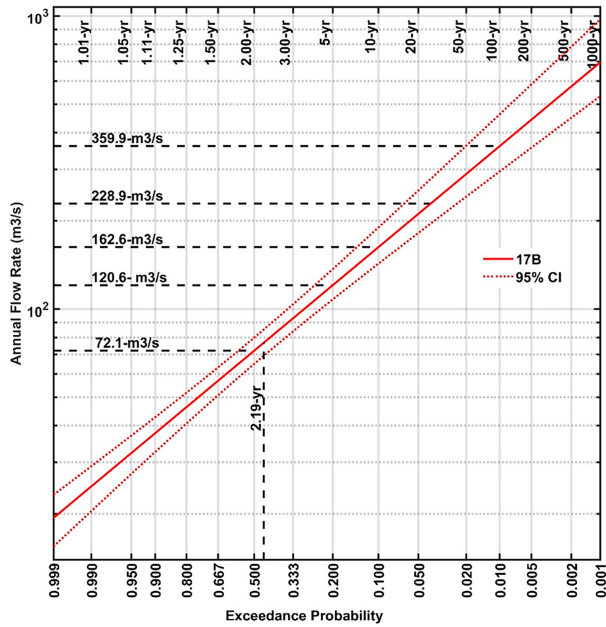


FIG. 5. Result of flood frequency estimates obtained from LPT III distributions as shown for USGS gauge station 01181000 (West Branch Westfield River at Huntington, Massachusetts).

In the RF modeling process, the 100-yr flood flow is considered a response variable (hereafter named “observed streamflow”).

b. Model evaluation

To examine the accuracy of the RF model, a k -fold validation experiment with 5 folds was performed using the observed streamflow (LPT-III-based flood estimates) dataset for the 98 gauge stations. The scatterplot of RF-predicted streamflow and observed streamflow was produced (Fig. 6). Using the RF model, a high magnitude of R^2 (0.82) was obtained for the study regions. Though this suggests a high correlation, the very extreme values ($>1500 \text{ m}^3 \text{ s}^{-1}$) clearly display higher errors than the other values. These extreme flows, which occur for the largest watersheds, likely display higher errors because daily precipitation is no longer the largest driver of the 100-yr flood for larger watersheds. This is supported in Fig. 9, where streamflow is displaying minimal dependence on the largest precipitation values ($>100 \text{ mm}$). Additionally, while the model exhibits higher errors for these large watersheds, these locations encompass long stream reaches and are more likely to contain streamgauges along the main stem or tributaries. Consequently, extreme flow estimates for watersheds of these sizes ($>400 \text{ mi}^2$ or $\sim 1000 \text{ km}^2$) are derived from observed data rather than regression models in practice.

Boxplots for the 100-yr flood estimates (RF and observed) for 98 gauge stations are presented to illustrate relative variability between RF-predicted and observed estimates (Fig. 7). These results suggest no substantial differences between these flood flow estimates. Specifically, the boxplots indicate both flood estimates (RF and observed datasets) are nearly equally distributed around the median (central mark) along with

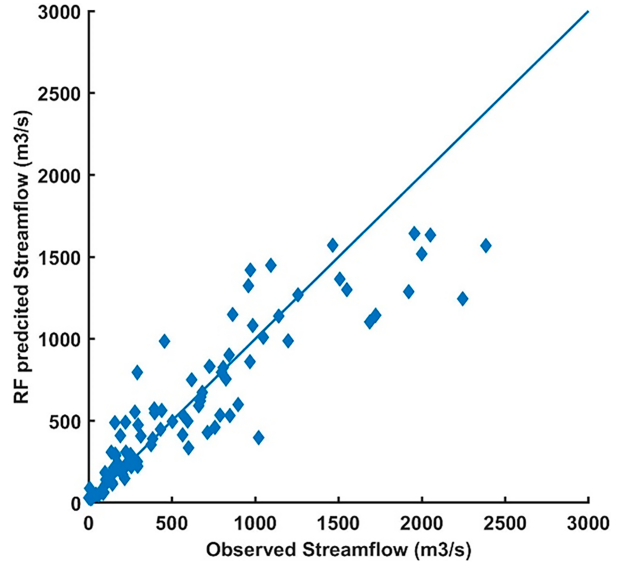


FIG. 6. Observed (LPT-III-based estimates) and model-predicted (RF-based estimates) 100-yr flood flow magnitudes at 98 streamgauges used in the flood flow analysis for the Northeast United States.

flood quantiles (first and third quartiles). As anticipated, RF and observed datasets have the largest accuracy range, indicating their accuracy for estimation. Additionally, Fig. 7 indicates that the RF underestimated the extreme flows for approximately 10% of the gauge stations. Generally, nonparametric methods such as RF or quantile RF (QRF) models may not capture extreme (both low and high) values well due to the weakness of the empirical distribution function at modeling probabilities close to 0 (Bhuiyan et al. 2018; He et al. 2017). Moreover, these studies have shown that RF or QRF can perform better in generating one-sided prediction intervals.

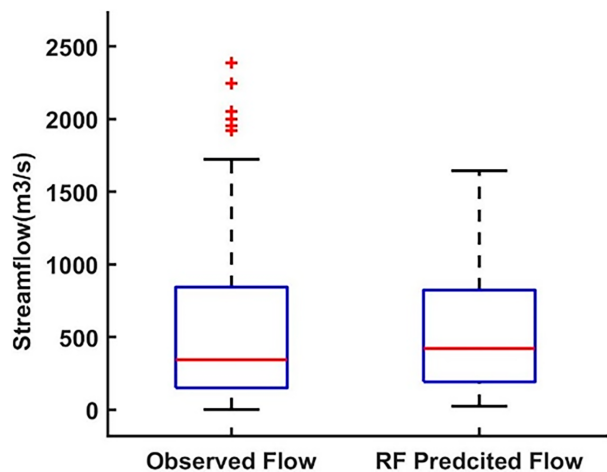


FIG. 7. Boxplot of observed (LPT-III-based flood estimates), and model-predicted (RF-based flood estimates) 100-yr flood flow magnitudes for 98 USGS gauge stations. In each box, the central mark is the median, and the edges are the first and third quartiles.

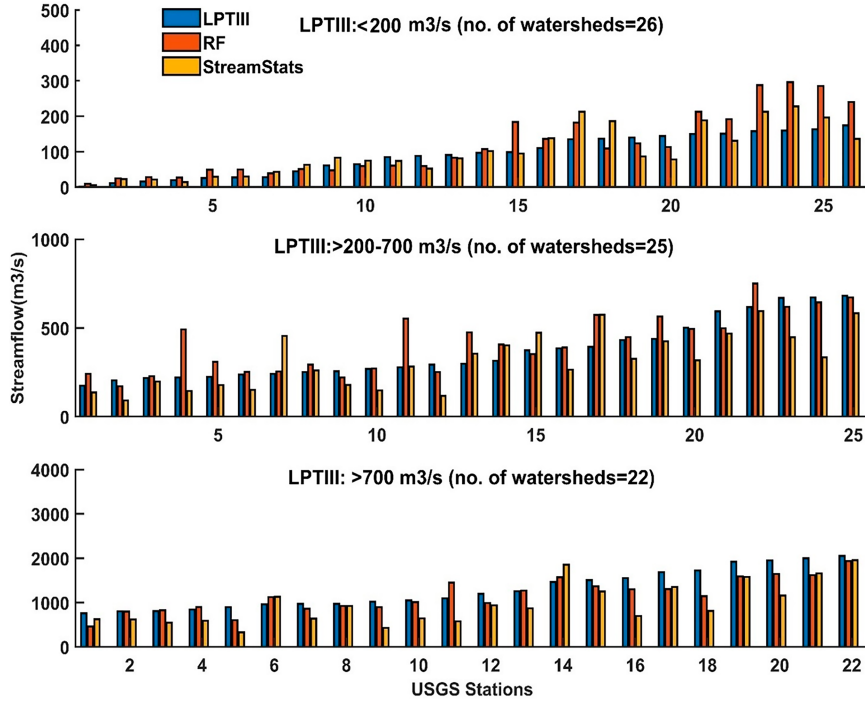


FIG. 8. Comparison among observed (LPT-III-based flood estimates), ML-model-predicted (RF-based flood estimates), and StreamStats-derived 100-yr flood flow magnitudes streamflow for 73 USGS gauge stations.

As an additional evaluation step, the observed (LPT-III-based flood estimates), ML-model-predicted (RF-based flood estimates), and StreamStats-derived 100-yr flood flow estimates for each USGS site number are compared (Fig. 8). StreamStats-derived flood flow estimates were available for 73 stream-gauges in the Northeast United States. Estimates generated by RF and StreamStats are consistent for smaller watersheds with 100-yr flows less than $200 \text{ m}^3 \text{ s}^{-1}$. Figure 8 demonstrates that the RF method generated significantly better flood flow estimates than the StreamStats estimates for 100-yr flows in the larger basins (greater than $700 \text{ m}^3 \text{ s}^{-1}$).

c. Error analysis

To compare the validity of the different models (StreamStats and RF), three indicators were evaluated: NCRMSE, MARE, and KGE (Table 1). The results revealed that the random error of the RF estimates was significantly lower (27%) than the StreamStats estimates (42%). In addition, StreamStats and RF estimates exhibited very low MARE values (33%–34%) over the study region.

These results indicate that the ML-based flood flow model was able to considerably reduce the random error for all the watersheds (27%). Similarly, RF-based estimates exhibited very low MARE values (34%) for the study sites. Moreover, the magnitude of KGE for the ML-based cross-validation technique was noticeably high (0.79) compared to the StreamStats approach. The results suggest that the ML technique could be applied to any ungauged locations to facilitate the better prediction of flood flow. Overall, the cross-validation experiments

showed the RF technique outperformed the traditional StreamStats-based regression techniques.

d. Sensitivity analysis

To demonstrate the potential insight into the influence of individual predictors on the response variable, a PDP (Friedman 2001; Crawford et al. 2019; Zhao and Hastie 2021) was derived for the input variables. PDPs provide a visual representation of the model's assessment of each predictor variable's contribution to estimating the response variable (e.g., the 100-yr flood). Unlike traditional regression models, which assign a single coefficient to quantify a variable's influence, machine learning models can capture nonlinear relationships and threshold effects. Specifically, the PDPs illustrate points at which a variable's predictive significance changes. The results in Fig. 9 suggest that precipitation, temperature, area, and elevation have significant variability in their significance for flood flow prediction. The PDP test was applied for the remaining variables separately to identify the contribution information in the modeling process. Similarly, other indicators also characterize the key predictors

TABLE 1. Results of error metrics (KGE, NCRMSE, and MARE) are presented for the 73 gauge stations.

Error metrics	StreamStats	RF
NCRMSE	0.42	0.27
MARE	0.33	0.34
KGE	0.70	0.79

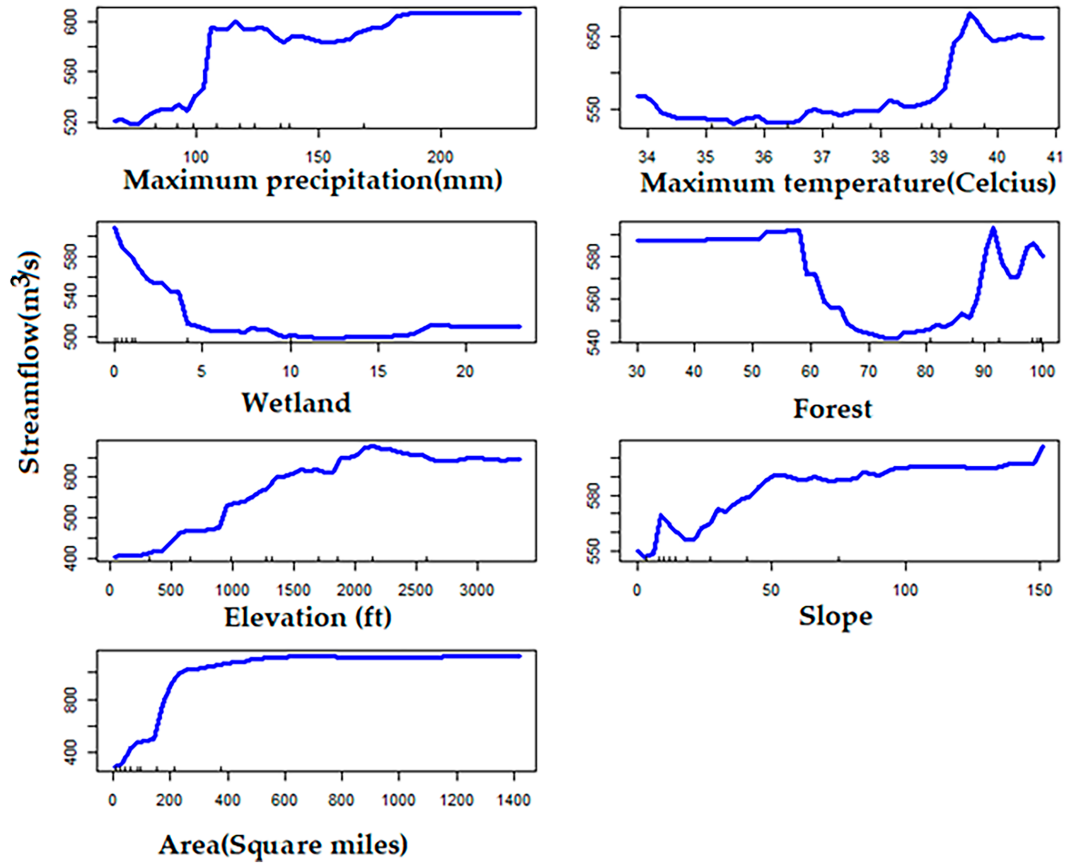


FIG. 9. PDP for the input variables used in this study.

by producing considerable impact in predicting flood flows for the study area. Overall, Fig. 9 demonstrates that all variables selected have a considerable impact on the model prediction, justifying the choice for including them in the final model.

The RF algorithm utilizes a tree-based ensemble method which is well suited for high-dimensional data for the estimation of flood flow prediction. Therefore, RF successfully incorporated different predictor variables of mixed types, such as meteorological (“precipitation,” “maximum temperature”) and physical (“area,” “elevation,” and ‘land cover’) data, while also offering a robust way of blending multisource datasets, providing at the same time improved flood estimates and characterization of its uncertainty.

5. Summary and conclusions

Accurate estimation of the 100-yr flood flow is vital for hydroclimatic applications, making it important for both regional and global studies. In the Northeast and throughout the United States in general, many river basins are not gauged adequately, making estimates of 100-yr flood events difficult and requiring managers to rely on streamflow data from nearby basins. Therefore, this study investigated the use of a nonparametric ML algorithm for advancing flood flow prediction techniques for combining multiple meteorological variables along with

other auxiliary variables (slope, watershed area, land cover, and elevation) in ungauged basins in the Northeast United States. Understanding complex hydrological processes is fundamental to creating accurate estimates of extreme events, and employing new ML techniques can be used to increase understanding of these extreme phenomena.

The error metrics used in this study are normalized centered root-mean-square error (NCRMSE), mean absolute relative error (MARE), and Kling–Gupta efficiency (KGE) for the assessment of flood models. The ML algorithm improved the flood prediction by reducing significant systematic and random errors for LPT-III extrapolation estimates. Furthermore, KGE provided a more generalized satisfactory skill score for assessing the accuracy of the flood predictions relative to the random chance. One of the challenges in utilizing a data-driven machine learning algorithm is overfitting. However, the RF model presented here is tested using a robust k -fold validation experiment, suggesting that it is directly applicable to other similar hydroclimatic regions for which the model has not been calibrated. Figure 4 demonstrates that the RF model error stabilized (converged) by selecting the appropriate values of hyperparameters. The flood prediction is designed to assess the sensitivity of the blending technique to the different forcing variables. Additionally, a partial dependency analysis was conducted over the study area, and the results demonstrated that

all features are comparatively important by showing a high level of significance for all of the variables applied. This sensitivity analysis also demonstrates the best input features for 100-yr flood prediction using a machine learning algorithm for the modeling process.

This analysis supports that the proposed flood prediction methodology is robust and accurate given its promising performance in a large study area with heterogeneity in terrain morphology and extreme precipitation. Thus, it can be applied to other complex terrain regions, where high-resolution precipitation data are available. Future studies could include increasing the number of study sites for different hydroclimatic regions, investigating the processes dictating the threshold where flood models such as this one begin to perform poorly, or focusing on other land surface variables (such as soil moisture) and meteorological conditions (such as cloud properties) to improve prediction.

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Data availability statement. The data supporting this paper can all be found at the author's Zenodo (<https://doi.org/10.5281/zenodo.7082676>).

REFERENCES

- Abraham, S., C. Huynh, and H. Vu, 2019: Classification of soils into hydrologic groups using machine learning. *Data*, **5**, 2, <https://doi.org/10.3390/data5010002>.
- Ahmad, T., A. C. Pandey, and A. Kumar, 2019: Evaluating urban growth and its implication on flood hazard and vulnerability in Srinagar city, Kashmir Valley, using geoinformatics. *Arabian J. Geosci.*, **12**, 308, <https://doi.org/10.1007/s12517-019-4458-1>.
- Balica, S. F., I. Popescu, L. Beevers, and N. G. Wright, 2013: Parametric and physically based modelling techniques for flood risk and vulnerability assessment: A comparison. *Environ. Modell. Software*, **41**, 84–92, <https://doi.org/10.1016/j.envsoft.2012.11.002>.
- Beck, H. E., A. I. J. M. van Dijk, A. De Roo, D. G. Miralles, T. R. McVicar, J. Schellekens, and L. A. Bruijnzeel, 2016: Global-scale regionalization of hydrologic model parameters. *Water Resour. Res.*, **52**, 3599–3622, <https://doi.org/10.1002/2015WR018247>.
- Bergstra, J., and Y. Bengio, 2012: Random search for hyper-parameter optimization. *J. Mach. Learn. Res.*, **13**, 281–305, <https://jmlr.csail.mit.edu/papers/volume13/bergstra12a/bergstra12a.pdf>.
- Bhuiyan, M. A. E., E. I. Nikolopoulos, E. N. Anagnostou, P. Quintana-Seguí, and A. Barella-Ortiz, 2018: A non-parametric statistical technique for combining global precipitation datasets: Development and hydrological evaluation over the Iberian Peninsula. *Hydrol. Earth Syst. Sci.*, **22**, 1371–1389, <https://doi.org/10.5194/hess-22-1371-2018>.
- , —, and —, 2019: Machine learning-based blending of satellite and reanalysis precipitation datasets: A multiregional tropical complex terrain evaluation. *J. Hydrometeorol.*, **20**, 2147–2161, <https://doi.org/10.1175/JHM-D-19-0073.1>.
- Blöschl, G., C. Reszler, and J. Komma, 2008: A spatially distributed flash flood forecasting model. *Environ. Modell. Software*, **23**, 464–478, <https://doi.org/10.1016/j.envsoft.2007.06.010>.
- Bobee, B., 1975: The Log Pearson Type 3 distribution and its application in hydrology. *Water Resour. Res.*, **11**, 681–689, <https://doi.org/10.1029/WR011i005p00681>.
- Breiman, L., 2001: Random forests. *Mach. Learn.*, **45**, 5–32, <https://doi.org/10.1023/A:1010933404324>.
- Burkey, J., 2022: Log-Pearson flood flow frequency using USGS 17B. MATLAB Central File Exchange, accessed 18 March 2022, <https://www.mathworks.com/matlabcentral/fileexchange/22628-log-pearson-flood-flow-frequency-using-usgs-17b>.
- Carsell, K. M., N. D. Pingel, and D. T. Ford, 2004: Quantifying the benefit of a flood warning system. *Nat. Hazards Rev.*, **5**, 131–140, [https://doi.org/10.1061/\(ASCE\)1527-6988\(2004\)5:3\(131\)](https://doi.org/10.1061/(ASCE)1527-6988(2004)5:3(131)).
- Cayan, D. R., K. T. Redmond, and L. G. Riddle, 1999: ENSO and hydrologic extremes in the western United States. *J. Climate*, **12**, 2881–2893, [https://doi.org/10.1175/1520-0442\(1999\)012<2881:EAHEIT>2.0.CO;2](https://doi.org/10.1175/1520-0442(1999)012<2881:EAHEIT>2.0.CO;2).
- Chakraborty, J., G. A. Tobin, and B. E. Montz, 2005: Population evacuation: Assessing spatial variability in geophysical risk and social vulnerability to natural hazards. *Nat. Hazards Rev.*, **6**, 23–33, [https://doi.org/10.1061/\(ASCE\)1527-6988\(2005\)6:1\(23\)](https://doi.org/10.1061/(ASCE)1527-6988(2005)6:1(23)).
- Chen, W., X. Zhao, P. Tsangaratos, H. Shahabi, I. Ilia, W. Xue, X. Wang, and B. B. Ahmad, 2020: Evaluating the usage of tree-based ensemble methods in groundwater spring potential mapping. *J. Hydrol.*, **583**, 124602, <https://doi.org/10.1016/j.jhydrol.2020.124602>.
- Chiang, Y.-M., K.-L. Hsu, F.-J. Chang, Y. Hong, and S. Sorooshian, 2007: Merging multiple precipitation sources for flash flood forecasting. *J. Hydrol.*, **340**, 183–196, <https://doi.org/10.1016/j.jhydrol.2007.04.007>.
- Coleman, T., M. F. Dorn, K. Kaufeld, and L. Mentch, 2021: Forecasting hurricane-related power outages via locally optimized random forests. *Stat.*, **10**, e346, <https://doi.org/10.1002/sta4.346>.
- Crawford, J., K. Venkataraman, and J. Booth, 2019: Developing climate model ensembles: A comparative case study. *J. Hydrol.*, **568**, 160–173, <https://doi.org/10.1016/j.jhydrol.2018.10.054>.
- DelSanto, A., M. A. E. Bhuiyan, K. A. Andreadis, and R. N. Palmer, 2023: Low-flow (7-day, 10-year) classical statistical and improved machine learning estimation methodologies. *Water*, **15**, 2813, <https://doi.org/10.3390/w15152813>.
- Devi, N. N., B. Sridharan, and S. N. Kuiry, 2019: Impact of urban sprawl on future flooding in Chennai city, India. *J. Hydrol.*, **574**, 486–496, <https://doi.org/10.1016/j.jhydrol.2019.04.041>.
- Doocy, S., A. Daniels, S. Murray, and T. D. Kirsch, 2013: The human impact of floods: A historical review of events 1980–2009 and systematic literature review. *PLOS Curr.*, **5**, <https://doi.org/10.1371/currents.dis.f4deb457904936b07c09daa98ee8171a>.
- Formetta, G., R. Mantilla, S. Franceschi, A. Antonello, and R. Rigon, 2011: The JGrass-NewAge system for forecasting and managing the hydrological budgets at the basin scale: Models of flow generation and propagation/routing. *Geosci. Model Dev.*, **4**, 943–955, <https://doi.org/10.5194/gmd-4-943-2011>.
- Friedman, J. H., 2001: Greedy function approximation: A gradient boosting machine. *Ann. Stat.*, **29**, 1189–1232, <https://doi.org/10.1214/aos/1013203451>.
- Griffis, V. W., and J. R. Stedinger, 2007: Log-Pearson Type 3 distribution and its application in flood frequency analysis. I: Distribution characteristics. *J. Hydrol. Eng.*, **12**, 482–491, [https://doi.org/10.1061/\(ASCE\)1084-0699\(2007\)12:5\(482\)](https://doi.org/10.1061/(ASCE)1084-0699(2007)12:5(482)).

- Gupta, H. V., Bastidas, L. A. Sorooshian, S. Shuttleworth, W. J. Yang, Z. L. N D H. Kling, K. K. Yilmaz, and G. F. Martinez, 2009: Decomposition of the mean squared error and NSE performance criteria: Implications for improving hydrological modelling. *J. Hydrol.*, **377**, 80–91, <https://doi.org/10.1016/j.jhydrol.2009.08.003>.
- Guthrie, J. D., C. Dartiguenave, and K. G. Ries III, 2009: Web services in the U.S. Geological Survey StreamStats web application. 2009 *Int. Conf. on Advanced Geographic Information Systems & Web Services*, Cancun, Mexico, Institute of Electrical and Electronics Engineers, 60–63, <https://doi.org/10.1109/GEOWS.2009.36>.
- Haddad, K., and A. Rahman, 2011: Selection of the best fit flood frequency distribution and parameter estimation procedure: A case study for Tasmania in Australia. *Stochastic Environ. Res. Risk Assess.*, **25**, 415–428, <https://doi.org/10.1007/s00477-010-0412-1>.
- Hapuarachchi, H. A. P., Q. J. Wang, and T. C. Pagano, 2011: A review of advances in flash flood forecasting. *Hydrol. Processes*, **25**, 2771–2784, <https://doi.org/10.1002/hyp.8040>.
- He, J., D. W. Wanik, B. M. Hartman, E. N. Anagnostou, M. Astitha, and M. E. B. Frediani, 2017: Nonparametric tree-based predictive modeling of storm outages on an electric distribution network. *Risk Anal.*, **37**, 441–458, <https://doi.org/10.1111/risa.12652>.
- Hu, L., E. I. Nikolopoulos, F. Marra, and E. N. Anagnostou, 2020: Sensitivity of flood frequency analysis to data record, statistical model, and parameter estimation methods: An evaluation over the contiguous United States. *J. Flood Risk Manage.*, **13**, e12580, <https://doi.org/10.1111/jfr3.12580>.
- Hunt, J. C. R., 2005: Inland and coastal flooding: Developments in prediction and prevention. *Philos. Trans. Roy. Soc.*, **A363**, 1475–1491, <https://doi.org/10.1098/rsta.2005.1580>.
- Jessup, S. M., and S. J. Colucci, 2012: Organization of flash-flood-producing precipitation in the Northeast United States. *Wea. Forecasting*, **27**, 345–361, <https://doi.org/10.1175/WAF-D-11-00026.1>.
- Karran, D. J., E. Morin, and J. Adamowski, 2014: Multi-step streamflow forecasting using data-driven non-linear methods in contrasting climate regimes. *J. Hydroinf.*, **16**, 671–689, <https://doi.org/10.2166/hydro.2013.042>.
- Khosravi, K., A. Golkarian, M. J. Booi, R. Barzegar, W. Sun, Z. M. Yaseen, and A. Mosavi, 2021: Improving daily stochastic streamflow prediction: Comparison of novel hybrid data-mining algorithms. *Hydrol. Sci. J.*, **66**, 1457–1474, <https://doi.org/10.1080/02626667.2021.1928673>.
- Kratzert, F., D. Klotz, M. Herrnegger, A. K. Sampson, S. Hochreiter, and G. S. Nearing, 2019a: Toward improved predictions in ungauged basins: Exploiting the power of machine learning. *Water Resour. Res.*, **55**, 11 344–11 354, <https://doi.org/10.1029/2019WR026065>.
- , —, G. Shalev, G. Klambauer, S. Hochreiter, and G. Nearing, 2019b: Towards learning universal, regional, and local hydrological behaviors via machine learning applied to large-sample datasets. *Hydrol. Earth Syst. Sci.*, **23**, 5089–5110, <https://doi.org/10.5194/hess-23-5089-2019>.
- Li, J., Y. Chen, H. Wang, J. Qin, J. Li, and S. Chiao, 2017: Extending flood forecasting lead time in a large watershed by coupling WRF QPF with a distributed hydrological model. *Hydrol. Earth Syst. Sci.*, **21**, 1279–1294, <https://doi.org/10.5194/hess-21-1279-2017>.
- Lins, H. F., 2012: USGS hydro-climatic data network 2009 (HCDN–2009). USGS Fact Sheet 2012-3047, 4 pp., <https://pubs.usgs.gov/fs/2012/3047/>.
- Livneh, B., T. J. Bohn, D. W. Pierce, F. Munoz-Arriola, B. Nijssen, R. Vose, D. R. Cayan, and L. Brekke, 2015: A spatially comprehensive, hydrometeorological data set for Mexico, the U.S., and Southern Canada 1950–2013. *Sci. Data*, **2**, 150042, <https://doi.org/10.1038/sdata.2015.42>.
- , and NCAR Staff, Eds., 2019: The Climate Data Guide: Livneh gridded precipitation and other meteorological variables for continental US, Mexico and Southern Canada. Accessed 18 December 2020, <https://climatedataguide.ucar.edu/climate-data/livneh-gridded-precipitation-and-other-meteorological-variables-continental-us-mexico>.
- Meinshausen, N., and G. Ridgeway, 2006: Quantile regression forests. *J. Mach. Learn. Res.*, **7**, 983–999.
- Ming, X., Q. Liang, X. Xia, D. Li, and H. J. Fowler, 2020: Real-time flood forecasting based on a high-performance 2-D hydrodynamic model and numerical weather predictions. *Water Resour. Res.*, **56**, e2019WR025583, <https://doi.org/10.1029/2019WR025583>.
- Mosavi, A., P. Ozturk, and K. Chau, 2018: Flood prediction using machine learning models: Literature review. *Water*, **10**, 1536, <https://doi.org/10.3390/w10111536>.
- Nayak, P. C., K. P. Sudheer, and K. S. Ramasastri, 2005: Fuzzy computing based rainfall-runoff model for real time flood forecasting. *Hydrol. Processes*, **19**, 955–968, <https://doi.org/10.1002/hyp.5553>.
- Nearing, G. S., F. Kratzert, A. K. Sampson, C. S. Pelissier, D. Klotz, J. M. Frame, C. Prieto, and H. V. Gupta, 2021: What role does hydrological science play in the age of machine learning? *Water Resour. Res.*, **57**, e2020WR028091, <https://doi.org/10.1029/2020WR028091>.
- Pang, B., J. Yue, G. Zhao, and Z. Xu, 2017: Statistical downscaling of temperature with the random forest model. *Adv. Meteor.*, **2017**, 7265178, <https://doi.org/10.1155/2017/7265178>.
- Papacharalampous, G. A., and H. Tyralis, 2018: Evaluation of random forests and Prophet for daily streamflow forecasting. *Adv. Geosci.*, **45**, 201–208, <https://doi.org/10.5194/adgeo-45-201-2018>.
- Pfahl, S., P. A. O’Gorman, and E. M. Fischer, 2017: Understanding the regional pattern of projected future changes in extreme precipitation. *Nat. Climate Change*, **7**, 423–427, <https://doi.org/10.1038/nclimate3287>.
- Pham, L. T., L. Luo, and A. Finley, 2021: Evaluation of random forests for short-term daily streamflow forecasting in rainfall- and snowmelt-driven watersheds. *Hydrol. Earth Syst. Sci.*, **25**, 2997–3015, <https://doi.org/10.5194/hess-25-2997-2021>.
- Phine, H. N., and M. A. Hira, 1983: Log Pearson Type-3 distribution: Parameter estimation. *J. Hydrol.*, **64**, 25–37, [https://doi.org/10.1016/0022-1694\(83\)90058-6](https://doi.org/10.1016/0022-1694(83)90058-6).
- Ries, K. G., III, J. D. Guthrie, A. H. Rea, P. A. Steeves, and D. W. Stewart, 2008: StreamStats: A water resources web application. USGS Fact Sheet FS 2008–3067, 6 pp., <https://pubs.usgs.gov/fs/2008/3067/pdf/fs-2008-3067-508.pdf>.
- Saber, M., K. I. Abdrabo, O. M. Habiba, S. A. Kantosh, and T. Sumi, 2020: Impacts of triple factors on flash flood vulnerability in Egypt: Urban growth, extreme climate, and mismanagement. *Geosciences*, **10**, 24, <https://doi.org/10.3390/geosciences10010024>.
- Saleh, F., V. Ramaswamy, Y. Wang, N. Georgas, A. Blumberg, and J. Pullen, 2017: A multi-scale ensemble-based framework for forecasting compound coastal-riverine flooding: The Hackensack-

- Passaic watershed and Newark Bay. *Adv. Water Resour.*, **110**, 371–386, <https://doi.org/10.1016/j.advwatres.2017.10.026>.
- Scornet, E., 2017: Tuning parameters in random forests. *ESAIM*, **60**, 144–162, <https://doi.org/10.1051/proc/201760144>.
- Seaber, P. R., F. P. Kapinos, and G. L. Knapp, 1987: Hydrologic unit maps. USGS Water-Supply Paper 2294, 66 pp., <https://doi.org/10.3133/wsp2294>.
- Soleymani, S. A., S. Goudarzi, M. H. Anisi, W. H. Hassan, M. Y. I. Idris, S. Shamshirband, N. M. Noor, and I. Ahmedy, 2016: A novel method to water level prediction using RBF and FFA. *Water Resour. Manage.*, **30**, 3265–3283, <https://doi.org/10.1007/s11269-016-1347-1>.
- Tabari, H., 2020: Author correction: Climate change impact on flood and extreme precipitation increases with water availability. *Sci. Rep.*, **10**, 16 969–16 910, <https://doi.org/10.1038/s41598-020-70816-2>.
- Tongal, H., and M. J. Booij, 2018: Simulation and forecasting of streamflows using machine learning models coupled with base flow separation. *J. Hydrol.*, **564**, 266–282, <https://doi.org/10.1016/j.jhydrol.2018.07.004>.
- Wing, O. E. J., and Coauthors, 2022: Inequitable patterns of US flood risk in the Anthropocene. *Nat. Climate Change*, **12**, 156–162, <https://doi.org/10.1038/s41558-021-01265-6>.
- Zhang, S., L. Lu, J. Yu, and H. Zhou, 2016: Short-term water level prediction using different artificial intelligent models. *2016 Fifth Int. Conf. on Agro-Geoinformatics (Agro-Geoinformatics)*, Tianjin, China, Institute of Electrical and Electronics Engineers, 1–6, <https://doi.org/10.1109/Agro-Geoinformatics.2016.7577678>.
- Zhao, Q., and T. Hastie, 2021: Causal interpretations of black-box models. *J. Bus. Econ. Stat.*, **39**, 272–281, <https://doi.org/10.1080/07350015.2019.1624293>.